

SARDAR PATEL UNIVERSITY

B.C.A Semester – II (Reg. & NC)

(CBCS) June - 2011 BATCH Onwards

Thursday, Date: 30th March 2017

Session: Morning Time : 10:00 A.M. to 1:00 P.M.

Subject Code: US02CBCA01

Subject Title : Advanced 'C' Programming and Introduction to Data Structures

Total Marks: 70

Q1. Multiple Choice Questions. (Attempt all)

[10]

1. Which operator is used declare pointer variable.
A. Selection (->). C. Assignment (=)
B. Indirection (*) D. Dot (.)
2. Which of the following can be used to create a new type that can be used anywhere a type is permitted?
A. union C. typedef
B. pointer D. struct
3. Which of the following allows a portion of memory to be shared by different types of data?
A. Union C. File
B. Structure D. Array
4. To open file only for **reading** purpose, _____ mode is specified in fopen() function.
A. r C. a
B. rw D. w
5. Which of the following data structure store all the elements of same data type?
A. Structure C. Union
B. File D. Array
6. Which of the following is not the operation of stack?
A. Push C. Peep
B. Pop D. Delete
7. In Stack , elements are inserted & deleted from _____.
A. TOP C. Both A and B
B. BOTTOM D. None of these.
8. A linked list in which last node pointing to the first node is known as _____.
A. Singly linked list C. Doubly Linked list
B. Circular linked list D. None of the above
9. Data structure in which insertion and deletion of an elements occurs at opposite the end is known as _____.
A. Stack C. Queue
B. Array D. Linked List
10. Front and Rear pointers are related to _____ data structure.
A. Linked list C. Stack
B. Array D. Queue

(1)

(P.T.O)

- Q2. Answer the following short questions (Attempt any TEN) [20]**
1. List different pointer declaration style. Which one is preferable?
 2. List benefits of pointers.
 3. Define a structure called "student" consisting of integer members called weight and height, character string called name. Declare structure variable called stud1 along with definition.
 4. State purpose of getw() & putw() functions.
 5. What is dynamic memory allocation?
 6. State different file modes available to manage the files in C.
 7. State various applications of Stack data structure.
 8. List the examples of Non-Primitive Data Structures.
 9. Draw the representation of Stack data structure. What is TOP?
 10. State various types of Queue.
 11. Explain structure of typical node of a linked list with diagram.
 12. State various applications of Queue.
- Q3. a. Define pointer variable. Explain how to declare and initialize pointer variable? [05]**
Also explain how to access value of variable through pointer type variable?
- b. Write a note on Dynamic memory allocation. [05]**
- OR**
- Q3. a. Explain the concept of pointer-to-pointer with example. [05]**
- b. Explain valid & invalid operations of pointer arithmetic. [05]**
- Q4.a. What is Structure? Explain general form of structure definition, declaration of structure type variable and accessing structure members with suitable example. [06]**
- b. Explain the purpose, syntax and example of functions getc() and putc(). [04]**
- OR**
- Q4.a. Differentiate between: Structure and Union. [06]**
- b. Explain the purpose, syntax and example of functions fopen() and fclose(). [04]**
- Q5.a. Write an algorithm for PUSH and POP operations on stack. [06]**
- b. What do you mean Linear Data Structure? List few examples and explain any one in brief. [04]**
- OR**
- Q5.a. Write an algorithm for PEEK and CHANGE operations on stack. [06]**
- b. Draw the Hierarchical Structure of Data Structure. List the main operations that can be performed on Data Structure with its use? [04]**
- Q6. What is Linked list ? Write an algorithm to insert an element at the beginning into a Singly linked list. [10]**
- OR**
- Q6. What is Queue ? Write an algorithm to insert an element in a Queue. [10]**